Octoverse: AI leads Python to top language as the number of global developers surges

In this year's Octoverse report, we study how public and open source activity on GitHub shows how AI is expanding as the global developer community surges in size.

GitHub Staff · @github

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32 minutes

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unprecedented number of developers join GitHub from across the globe, and many of these developers are contributing to open source projects for the first time.

In 2024, Python overtook JavaScript as the most popular language on GitHub, while Jupyter Notebooks skyrocketed—both of which underscore the surge in data science and machine learning on GitHub. We're also seeing increased interest in AI agents and smaller models that require less computational power, reflecting a shift across the industry as more people focus on new use cases for AI.

Our data also shows *a lot* more people are joining the global developer community. In the past year, more developers joined GitHub and engaged with open source and public projects (in some cases, empowered by AI). And since tools like GitHub Copilot started going mainstream in early 2023, the number of developers on GitHub has rapidly grown with significant gains in the global south. While we see signals that AI is driving interest in software development, we can't fully explain the surge in global growth our data reflects (but we'll keep studying it).

At GitHub, we know the critical role open source plays in bridging early experimentation and widespread adoption. In this year's Octoverse report, we'll explore how AI and a rapidly growing global developer community are coming together with compounding results.

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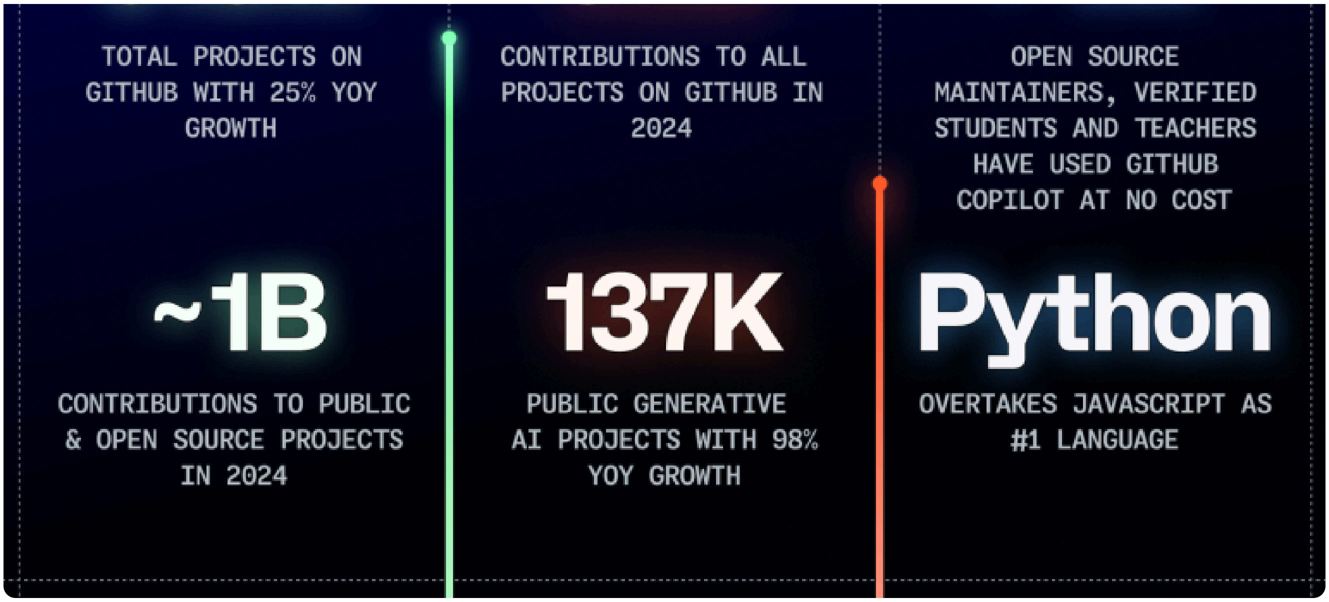

We uncover three big trends:

- **A surge in global generative AI activity**. AI is growing and evolving fast, and developers globally are going far beyond code generation with today's tools and models. While the United States leads in contributions to generative AI projects on GitHub, we see more absolute activity outside the United States. In 2024, there was a 59% surge in the number of contributions to generative AI projects on GitHub and a 98% increase in the number of projects overall—and many of those contributions came from places like India, Germany, Japan, and Singapore.
- **A rapidly growing number of developers worldwide—especially in Africa, Latin America, and Asia.** Notable growth is occurring in India, which is expected to have the world's largest developer population on GitHub by 2028, as well as across Africa and Latin America. We also see Brazil's developer community growing fast. Some of this is attributable to students. The GitHub Education program, for instance, has had more than 7 million verified participants. We've also seen 100% year-over-year growth among students, teachers, and open source maintainers adopting GitHub Copilot as part of our complimentary access program. This suggests AI isn't just helping more people learn to write code or build software faster—it's also attracting and helping more people become developers. First-time open source contributors continue to show wide-scale interest in AI projects. But we aren't seeing signs that AI has hurt open source with low-quality contributions.

used heavily across machine learning, data science, scientific computing, hobbyist, and home automation fields among others. The rise in Python usage correlates with large communities of people joining the open source community from across the STEM world rather than the traditional community of software developers. This year, we also saw a 92% spike in usage across Jupyter Notebooks. This could indicate people in data science, AI, machine learning, and academia increasingly use GitHub. Systems programming languages, like Rust, are also on the rise, even as Python, JavaScript, TypeScript, and Java remain the most widely used languages on GitHub.

💡 **Oh, and if you're a visual learner, we have you covered.** 👇

Octoverse 2024: The rise of Python and AI
GitHub

Watch on

A global community of developers that's growing fast

In early 2023, we celebrated reaching 100 million total developers on GitHub—and that number has climbed at a rapid rate since then. In 2024, developers around the world made more than **5.2 billion contributions to more than 518 million open source, public, and private projects**.

So, where in the world are GitHub developers most engaged, and where are we seeing the most growth? And as AI allows developers to code in the natural language of their choice, what parts of the world could we expect to see greater growth in? Let's take a look. 👇

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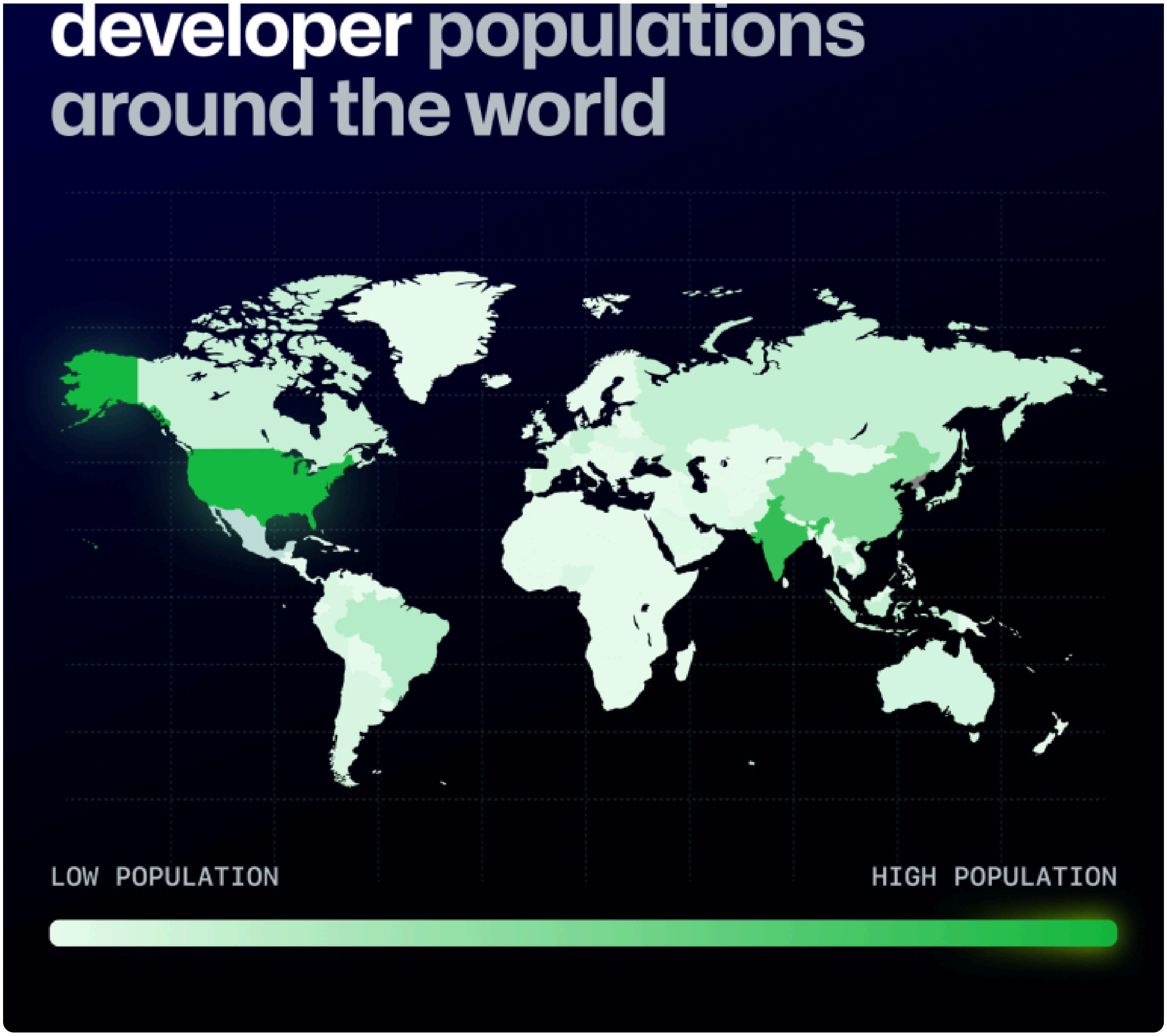

There's both stability and change among the top 20 countries with the highest number of developers on GitHub. While India continues to approach the number one spot (we now predict by 2028 based on updated projections, but more below), the United States continues to have the most developers worldwide on GitHub. Despite this, we have seen greater growth outside the United States every year since 2013—and that trend has sped up over the past few years.

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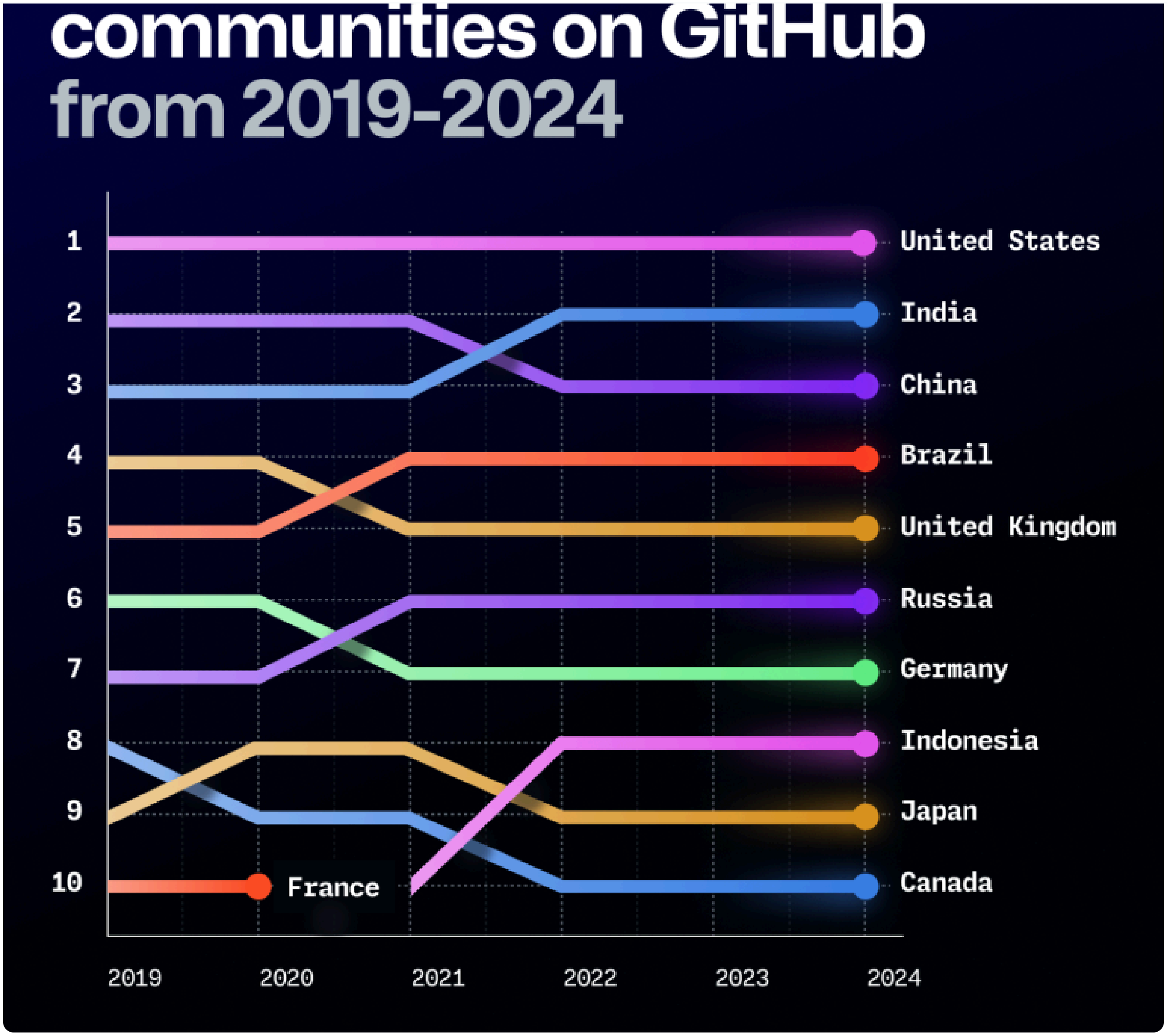


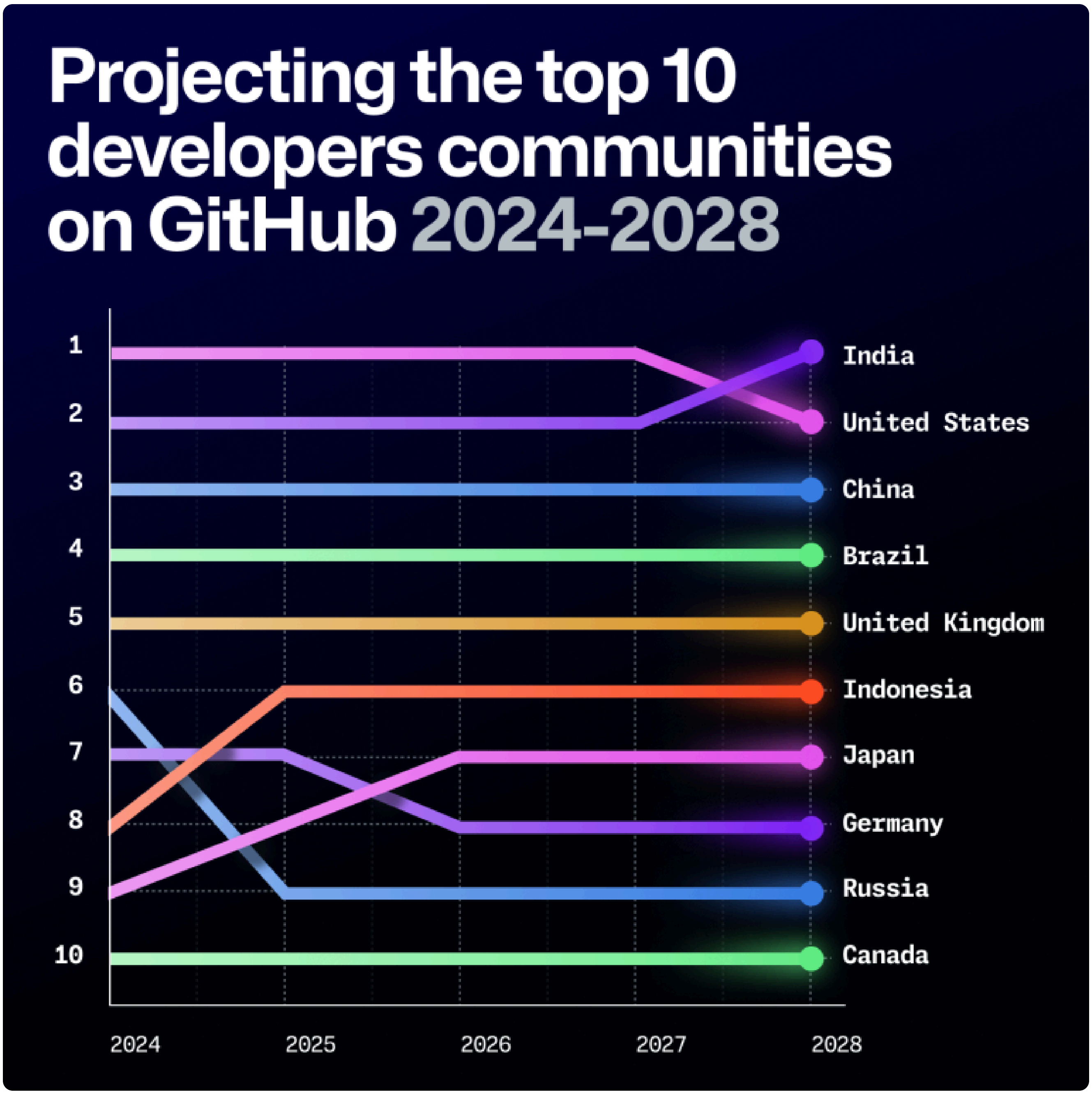
Globally, we see developer communities growing significantly. Brazil, India, and Nigeria are especially growing fast, which is notable given they are the most populous regions of their respective continents with linguistically diverse populations.

Overall, the top 20 developers communities have largely maintained their positions with a few notable exceptions. These include the Philippines (#18) overtaking Australia (#19) and Pakistan (#20) overtaking Poland (#21).

💡 **Stay smart.** The rise of these non-English, high-population regions is notable given that it is happening at the same time as the proliferation of generative AI tools, which are increasingly enabling developers to engage with code in their natural language.

/ Blog

To identify the developer communities with the highest growth potential over the next five years, we created projections based on current trends. And our prediction from last year changed: India is now on track to surpass the United States in number of developers on GitHub by 2028 (previously, we had predicted it would overtake the United States by 2027 based on linear population growth).



Fastest growing developer communities in Latin America

and governmental support that includes incentives for tech startups.

Country	Percentage growth	# of developers
Brazil	27% YoY	>5.4M
Mexico	21% YoY	>1.9M
Colombia	25% YoY	>1M
Argentina	22% YoY	>1.1M
Peru	27.5% YoY	>583K

Latin America spotlights

- **Peru** has seen a notable increase in fintech startups, and the country's tech sector is rapidly expanding due to foreign investments and a digital transformation accelerated by the pandemic.
- **Brazil's** open banking industry continues to be driven by Pix, the country's real-time payments infrastructure, for which the communication protocols were open sourced on GitHub by the Central Bank of Brazil. Brazil's government is also growing investment to attract private and public organizations to the country and recently announced a $4 billion proposal for an AI investment plan.
- **Mexico's** government has an independent software and developer ecosystem, and aims to boost the work force's coding and AI skills.

> "Students learn to collaborate and cooperate, they develop their soft skills. I did a survey at the end of the

technical and leadership skills." – **José Alfredo Román Cruz** // Professor, Technological Institute of Tlaxiaco

The fastest growing developer communities in Asia Pacific

The number of developers on GitHub in Asia Pacific communities is growing at some of the fastest rates globally—and we expect this trend to continue. This will be particularly true as generative AI increasingly empowers developers to engage with code and communities, regardless of their spoken language.

India has a fast growing developer community and is on pace to become the world's largest by 2028. India prioritizes open source software and introduced the National Education Policy of 2020, which requires schools to include coding and AI in student curriculum. And notably, a recent study from the learning platform Udemy found that GitHub is one of the most sought after skills in India, comparable to English grammar skills.

As part of the United Nations-backed Digital Public Goods Alliance, India also builds its digital public infrastructure with digital public goods (DPGs)—ranging from software code to AI models. Its Open Healthcare Network platform, for instance, is a community-driven project fueled by a small but dedicated team of open source developers who use GitHub Copilot.

Country	Percentage growth	# of developers
India	28% YoY	>17M
China	10% YoY	>9M
Indonesia	23% YoY	>3.5M

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Japan	23% YoY	>3.5M
Philippines	29% YoY	>1.7M

“Made in” collections

Want to know which open source projects are made in or receiving significant contributions from developers in a particular community? Check out the “Made in” collections on GitHub.

Asia Pacific spotlights

- **Singapore** has a high developer-to-population ratio, and government-led initiatives drive its tech sector, like its ambition to become a world-leading Smart Nation and its National AI Strategy.
- **Japan** has seen continued growth as its government promotes AI investments by offering a light-touch approach to regulation in the growing field.
- **South Korea**, a global IT leader with the fastest internet, focuses on AI, quantum computing, and cybersecurity. The country’s government is also investing heavily in technology and innovation as part of its K-Network 2030 strategy.

> “GitHub is like the air we breathe. It’s such a natural part of the way we work that sometimes we don’t even notice it. We cannot imagine living without GitHub.” – **Ryuzo Yamamoto** // Software Engineer, Souzoh

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surprise you 👀

In 2024, developer activity grew significantly in remote areas like Antarctica with 379% year-over-year growth, increasing from 19 to 91 developers on GitHub. Activity there is mostly focused on scientific research and marine science—and the growth we're seeing likely represents temporary population movement due to scientific research, rather than a long-term trend.

The fastest growing developer communities in Europe and the Middle East

Europe and the Middle East are shaping the future of AI in distinct ways. Countries in the Middle East are investing in AI with the aim to become global AI hubs. Meanwhile, over the last five years, the European Union has been putting forward several frameworks and laws to regulate technology and platform providers and generative AI, including the Digital Services Act, Digital Markets Act, AI Act, and Data Governance Act.

Country	Percentage growth	# of developers
United Kingdom	19% YoY	>4M
Germany	21% YoY	>3.5M
France	20% YoY	>2.8M
Spain	24% YoY	>1.8M
Turkey	19% YoY	>1.7M

Europe and Middle East spotlights

Spain's developer population is rising, securing the 10th largest GitHub user base globally. The country is committed to its plans to advance their national AI strategy by developing [Spanish language AI models].

- **France's** openness towards AI and tech innovation is also demonstrated through the country's [2030 plan], which includes investments in upskilling and attracting AI talent. Related is its effort to support start-up solutions through its [French Tech 2030 program].
- **Switzerland**, with just over 519,000 developers on GitHub, has taken a significant step to position itself at the forefront of public sector open source development by [mandating all federal software be open source] when possible. [The Economist also recently reported that Switzerland is the most innovative country], based in part on its contributions to open source projects on GitHub.
- **Turkey** has seen significant growth in its [information and communication technologies markets]. The country aims to enhance its banking, healthcare, and media sectors with 5G while relying on its local telecom operators to accelerate Internet of Things (IoT) and smart city projects.
- **The United Arab Emirates** is also a region to watch, as it recently committed to becoming a [global leader in AI and advanced technology] and saw a 32% year-over-year increase in developers on GitHub.

> "Basically, everything we build is iterating on open source. I think it's only reasonable for a larger organization like us to engage and give back to the community where possible." – **Kay Goebel //** Engineering Lead, [Zalando]

The fastest growing developer communities in Africa

Africa is nurturing an [increasing pool of developers] that is ready to drive the next wave of tech entrepreneurialism—and, in some cases, already are. The continent's developers have

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Country	Percentage growth	# of developers
Nigeria	28% YoY	>1.1M
Egypt	25% YoY	>990K
South Africa	23% YoY	>664K
Morocco	25% YoY	>556K
Kenya	33% YoY	>393K

Nigeria, Egypt, South Africa, and Kenya are considered Africa's "big four" countries focusing on technical literacy, drawing global investors, and securing most of Africa's startup funding in 2023. For example, **Egypt's** Ministry of Communication and Information Technology is developing technical skills through its "Our Future is Digital" initiative.

> "I think it's important to change the perception that Africans are merely consumers; we are creators as well. By helping people across Africa build projects and showcase their work globally, I hope to change this narrative and demonstrate that Africa is also a hub for innovation and creativity, especially within the open source community." – **Ruth Ikegah** // Community Lead, CHAOSS Africa

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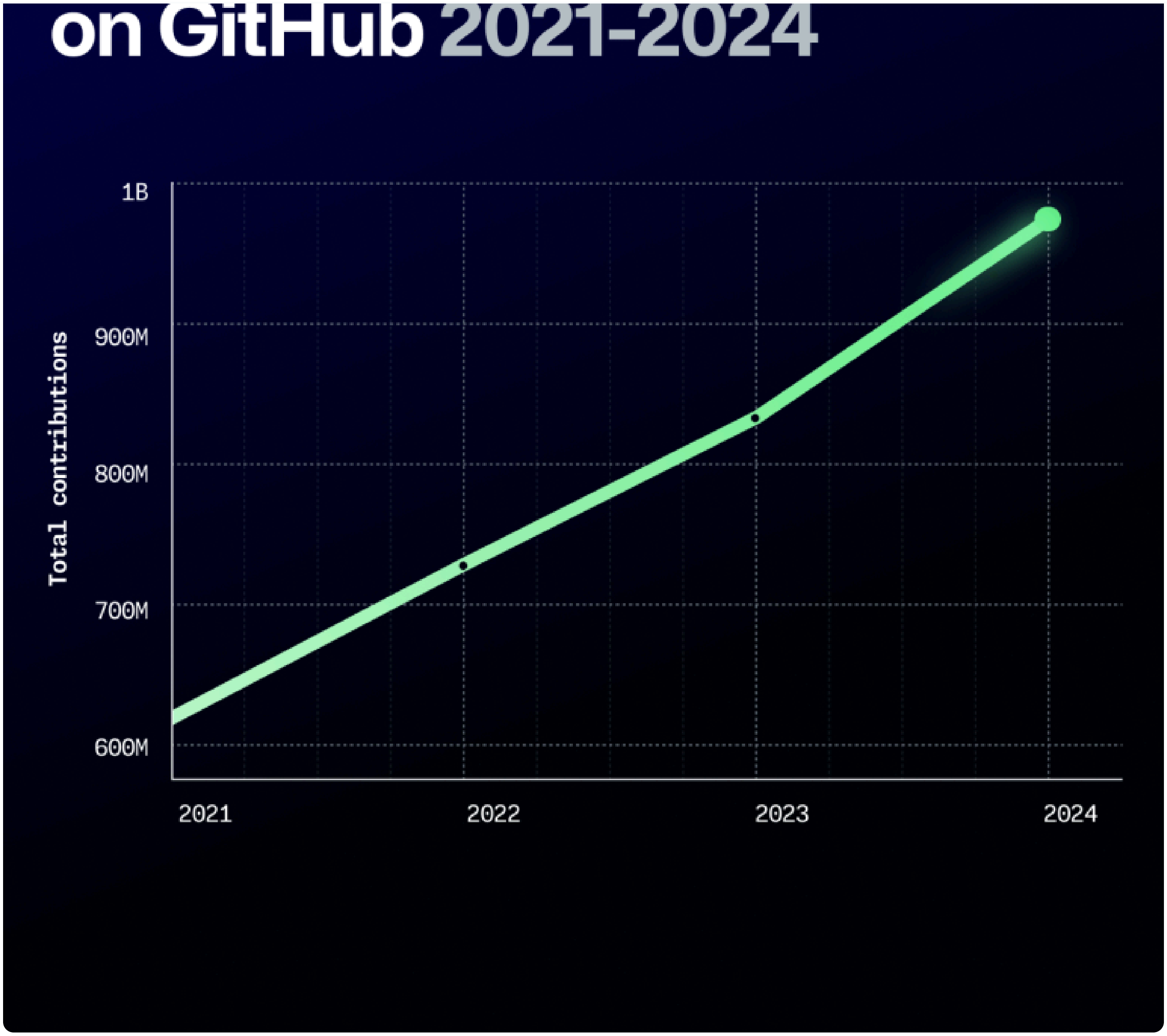
secondary schools.

- In 2019, **Nigeria** launched its "National Digital and Economy Policy and Strategy" to emphasize digital literacy as a cornerstone for economic development.
- **South Africa** also aims to train its young workforce in programming, AI, cloud computing, and robotics.
- **Morocco's** outsourcing was also cited as "a key sector" in advancing the country's digital transformation.

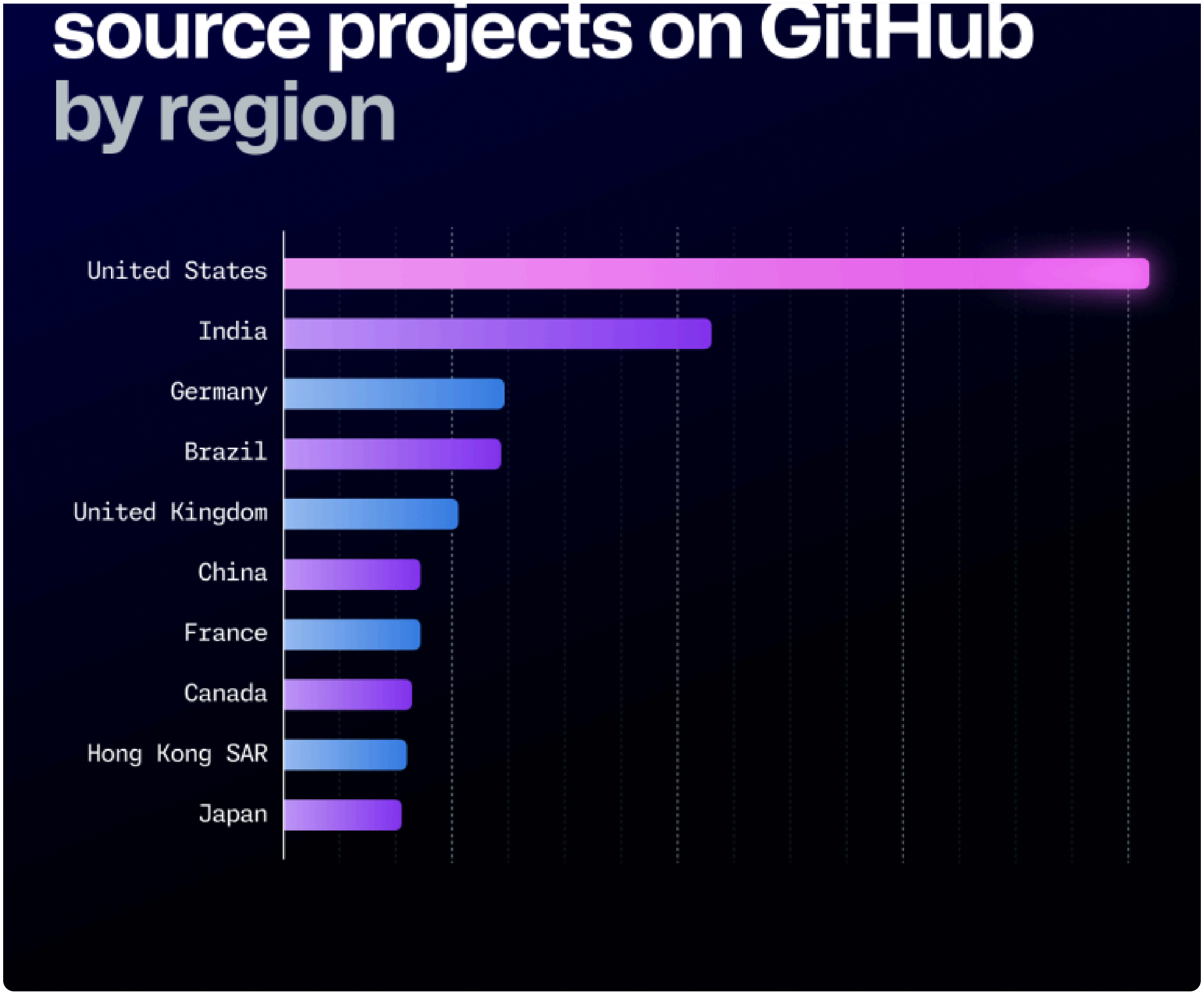
The state of open source

In 2024, developers globally made nearly **1 billion contributions** to open source and public repositories across GitHub (this includes open source projects with a license and public projects without a license accepted by the Open Source Initiative). These contributions ranged from popular projects like **home-assistant/core** to generative AI projects like **ollama/ollama** (more on that later) and commercially backed projects like **vercel/next.js**.

Similar to last year, we saw commercially backed and generative AI projects attract the most contributions in 2024. But where those contributions came from is notable with regions outside North America and Europe surging in overall activity.



More developers are consuming open source with a 15% spike in JavaScript packages through the npm registry. The top 50 packages saw net positive growth, which signals solidification and maturation of the JavaScript ecosystem. This also suggests that more people consume open source as key ecosystems—as with JavaScript—mature.



We see some differences between the top regions on GitHub and the top regions contributing to open source. For instance, Germany ranks as the third largest region contributing to open source on GitHub—but as of 2024, they are the seventh largest community on GitHub by developer population.

There's a continued increase in first-time contributors to open source projects. 1.4 million new developers globally joined open source with a majority contributing to commercially backed and generative AI projects. Notably, we did not see a rise in rejected pull requests. This could indicate that quality remains high despite the influx of new contributors.

Building more sustainable open source communities

Top 10 open source and public projects attracting the most first-time contributors in 2024 on GitHub	
1.	microsoft/vscode
2.	home-assistant/core
3.	microsoft/PowerToys
4.	Kas-tle/java2bedrock.sh
5.	ultralytics/ultralytics
6.	flutter/flutter
7.	langchain-ai/langchain
8.	Ultimaker/Cura
9.	platformio/platformio-home
10.	Koenkk/zigbee2mqtt

The top open source projects by contributors on GitHub. **home-assistant/core** and **flutter/flutter** continue to rank among the top projects by contributors on GitHub, reflecting their popularity and community strength. Notably, **vercel/next.js** showed up again in the top 10 list for all contributors, which indicates its continued growth and stature in web development.

exploring developing applications with large language models (LLMs) and cloud-native development practices.

- **ollama/ollama** also appeared in the top 10 public and open source projects, with the most contributions outranking large projects such as PyTorch and PowerToys. As of 2024, it's the third fastest-growing project by contributors on GitHub. This suggests a notable interest among developers in AI models that require less computational power (more on this later).
- Notably, the IoT project, **koenkk/zigbee2mqtt**, also emerged on our list this year—and that's likely due to the popularity of **home-assistant/core**, given **koenkk/zigbee2mqtt** can be used to get data such as room temperature. With **Ultimaker/Cura** on the list as well, it's clear that there is a large maker and hacking culture in open source.
- **One callout:** The programming language project, **ProvableHQ/leo**, which appeared for the first time in our top open source projects by contributors. As a statically typed language, Leo is often used for private applications developed on private, decentralized blockchain technologies.

Top 10 public projects by contributors on GitHub

Project	Contributor count
home-assistant/core	>21K
microsoft/vscode	>20K
ProvableHQ/leo	>20K
firstcontributions/first-contributions	>13k
flutter/flutter	>10K
NixOS/nixpkgs	>9K

/ Blog

langchain-ai/langchain	>8K
godotengine/godot	>7K
ollama/ollama	>7K

Developers are driving societal change through popular open source projects. GitHub's For Good First Issue is a curated list of DPGs that need contributors, connecting those projects with people who want to address a societal challenge and promote sustainable development. As more developers continue to join GitHub globally amid investments in connectivity and AI, we expect continued contribution growth in DPGs.

Top 10 For Good First Issue projects attracting first-time contributors in 2024
frappe/erpnext
truenas/charts
globaleaks/GlobaLeaks
ersilia-os/ersilia
ckan/ckan
learningequality/kolibri
ushahidi/platform
coronasafe/care

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rubyforgood/human-essentials
rubyforgood/casa

Significantly, 34% of contributors to the top 10 For Good Issue projects above made their first contribution after signing up for GitHub Copilot.

First-time contributors have contributed to projects that assist youth in the foster care system, facilitate drug discovery in middle- and low-income countries, enable whistleblowing around wrongdoings, and more.

More than 82% of GitHub contributions are made to private repositories. Developers made 4.3 billion contributions across more than 181 million private repositories in 2024. These numbers show the sheer scale of activity happening out of view in private repositories through free, Team, and GitHub Enterprise accounts—especially since we started offering private repositories to developers with free accounts in 2019.

A spike in Jupyter Notebooks use shows that open source underscores a growing community, especially as Python surges to become the most used language on GitHub. Since 2018, we have seen the use of Jupyter Notebooks steadily grow—and that growth surged in 2022 as research and experimentation with generative AI and machine learning took off. Since 2022, Jupyter Notebooks usage on GitHub has spiked more than 170%. And since last year, usage has increased by 92%. Data scientists and machine learning researchers commonly use the open source application for machine learning, data visualization, and more.

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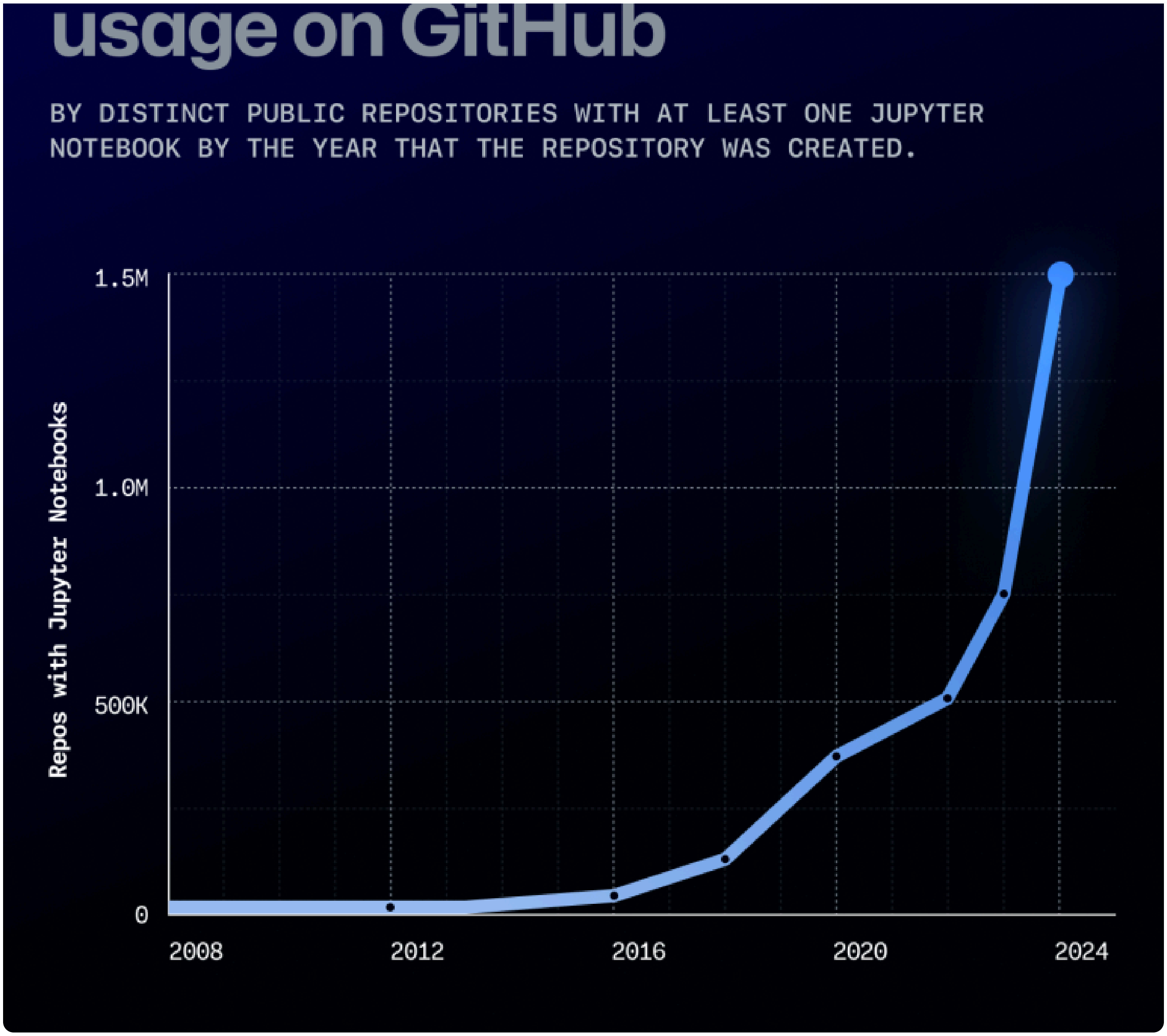


Explore GitHub's 2024 Open Source Survey: 3 key takeaways

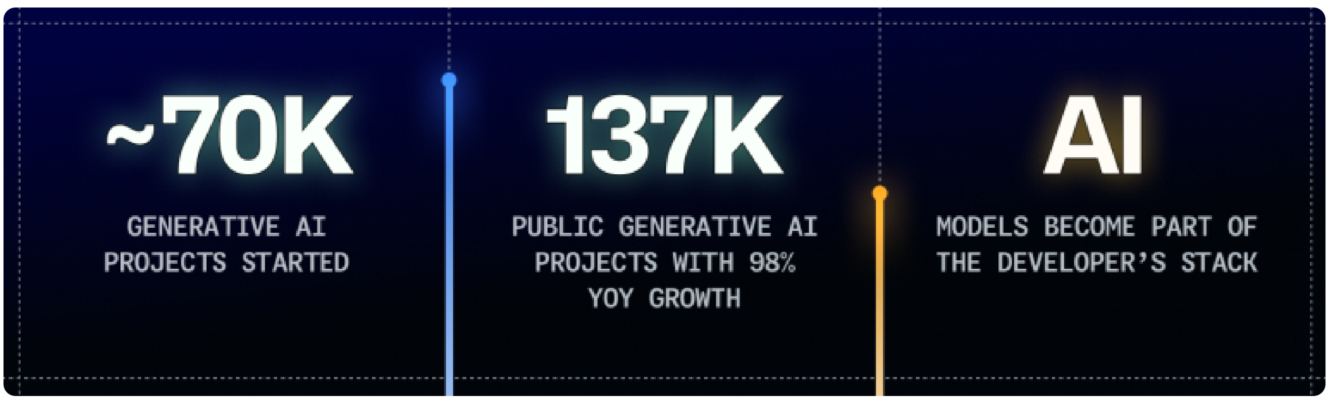
This year, we surveyed members of the open source community to gather data about the attitudes, experiences, and backgrounds of those who use, build, and maintain open source software. Here's some of what we found:

- **Security in open source is a priority.** Secure by design is gaining traction, with 82% of respondents considering it important to use an open source project, and 65% prioritizing it when contributing.

- **Open source is getting more diverse.** 30% of respondents classified themselves as minorities. This marks a nine percentage point or 43% increase since our last survey—and we anticipate an increasingly diverse open source community as the global developer community grows.

Explore the survey >

The state of generative AI in 2024



Over the past year, generative AI has moved beyond the hype of 2023 as developers and organizations alike look for results over experimentation—and data on GitHub shows as much. In 2024, developers on GitHub created over 70,000 **new public and open source generative AI projects** and made **almost 60% more total contributions** to all generative AI projects on GitHub.

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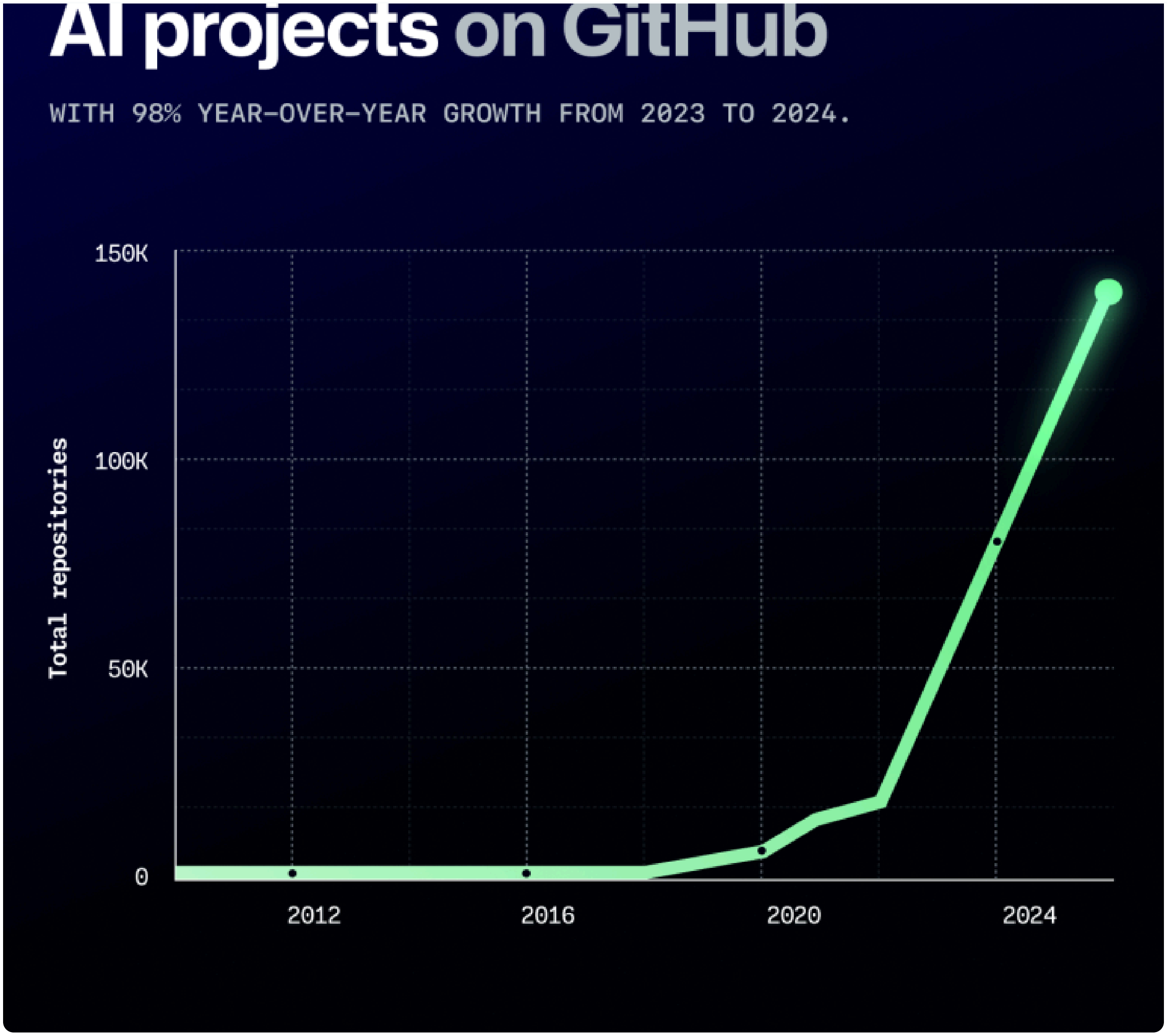

AI models become part of the developer's tech stack. We're seeing innovation in generative AI on GitHub move into public repositories, showing that developers are building more and more in the open. As developers identify more and more use cases for AI, the role of generative AI models in software development has shifted from helping developers write code to a new building block in developing applications.

- Yet, there's a growing need among developers for smaller models with good performance and lower compute costs, driven by a desire for the embedded use of AI models in smartphones.
- Notably, the fastest-growing open source AI project in 2024 by contributor count was [ollama/ollama](), suggesting increased experimentation with locally run LLMs.

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Build with AI on GitHub Models

At GitHub, we believe every developer can be an AI engineer with the right tools and training. From playground to coding with the model in GitHub Codespaces to production deployment via Azure, GitHub Models shows you how simple it can be.

[Sign up for the limited public beta >]

Developers on GitHub are trying to lower the barrier to AI experimentation. The top 10 public generative AI projects work to improve access to AI models to make experimentation easier. Applications range from creating user-friendly interfaces that improve text-to-image generation to building autonomous AI agents for task management. To pull this data, we looked for repositories that use generative AI-related keywords collected from our research last year.

Top 10 public generative AI projects in 2023 vs. 2024

	2023	**2024**
1	AUTOMATIC1111/stable-diffusion-webui	AUTOMATIC1111/stable-diffusion-webui
2	Significant-Gravitas/AutoGPT	Significant-Gravitas/AutoGPT
3	ChatGPTNextWeb/ChatGPT-Next-Web	ollama/ollama
4	Chanzhaoyu/chatgpt-web	nomic-ai/gpt4all
5	ggerganov/llama.cpp	binary-husky/gpt_academic
6	binary-husky/gpt_academic	comfyanonymous/ComfyUI

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8	lencx/ChatGPT	Chanzhaoyu/chatgpt-web
9	lm-sys/FastChat	chatchat-space/Langchain-Chatchat
10	huggingface/diffusers	open-webui/open-webui

While we see consistency in developer interest in image generation via AUTOMATIC1111/stable-diffusion-webui and AI agent development via Significant-Gravitas/AutoGPT, we also see some shifts in AI development work on GitHub:

- **A rise in smaller scale models.** In the last year, developers on GitHub have worked with Meta's LLaMA models, which suggests a growing interest in smaller, open source models.
- We also see via projects like **binary-husky/gpt_academic** a growing interest in developing AI tools for specialized use cases such as academic research.
- **A continued focus on developing AI agents to automate processes.** The continued presence of AutoGPT-related projects indicates that automation remains a significant area of exploration, with developers focusing on enhancing the capabilities of AI agents.

More than one million open source maintainers, and verified students and teachers have used GitHub Copilot at no cost. In 2024, we saw a 100% increase in teachers, students, and open source maintainers using GitHub Copilot in our complimentary program. This underscores AI's utility in education and upskilling (like learning a new programming language). In the last year, **over 450,000 GitHub Education users were first-time contributors** to projects on the platform.

💡 If you're a student, teacher, or maintainer, you can apply to get complimentary access to GitHub Copilot.

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five days a week and 8-15% among developers who use GitHub once a week) across open source and public projects. This echoes research conducted into AI coding tools' impact on overall perceived and quantitative productivity gains among developers.

We are already seeing growth in global contributions and contributors to generative AI projects. Developers in the **United States, Hong Kong SAR, India, Germany, and France** are among the top groups driving contributions to generative AI projects. India, for instance, had a 95% increase in year-over-year contributions to generative AI projects on GitHub while France had a 70% increase. These communities also saw some of the largest year-over-year growth in contributors.

Other communities saw some of the highest percentage growth in contributors to public generative AI projects, like the **Netherlands** (291%), **Ethiopia** (242%), **Costa Rica** (171%), **Serbia** (175%), and **Vietnam** (143%).

- These communities have fewer total contributors, causing any growth to result in a high percentage rate—but their growth still shows the global community of developers collaborating on generative AI projects.
- We anticipate this growth will continue, especially as more small language models are introduced in the broader marketplace, reducing computational requirements around developing software with AI. Moreover, as generative AI coding tools enable developers to write code with natural language, we see more opportunities for developers globally to contribute to projects, regardless of their native language.

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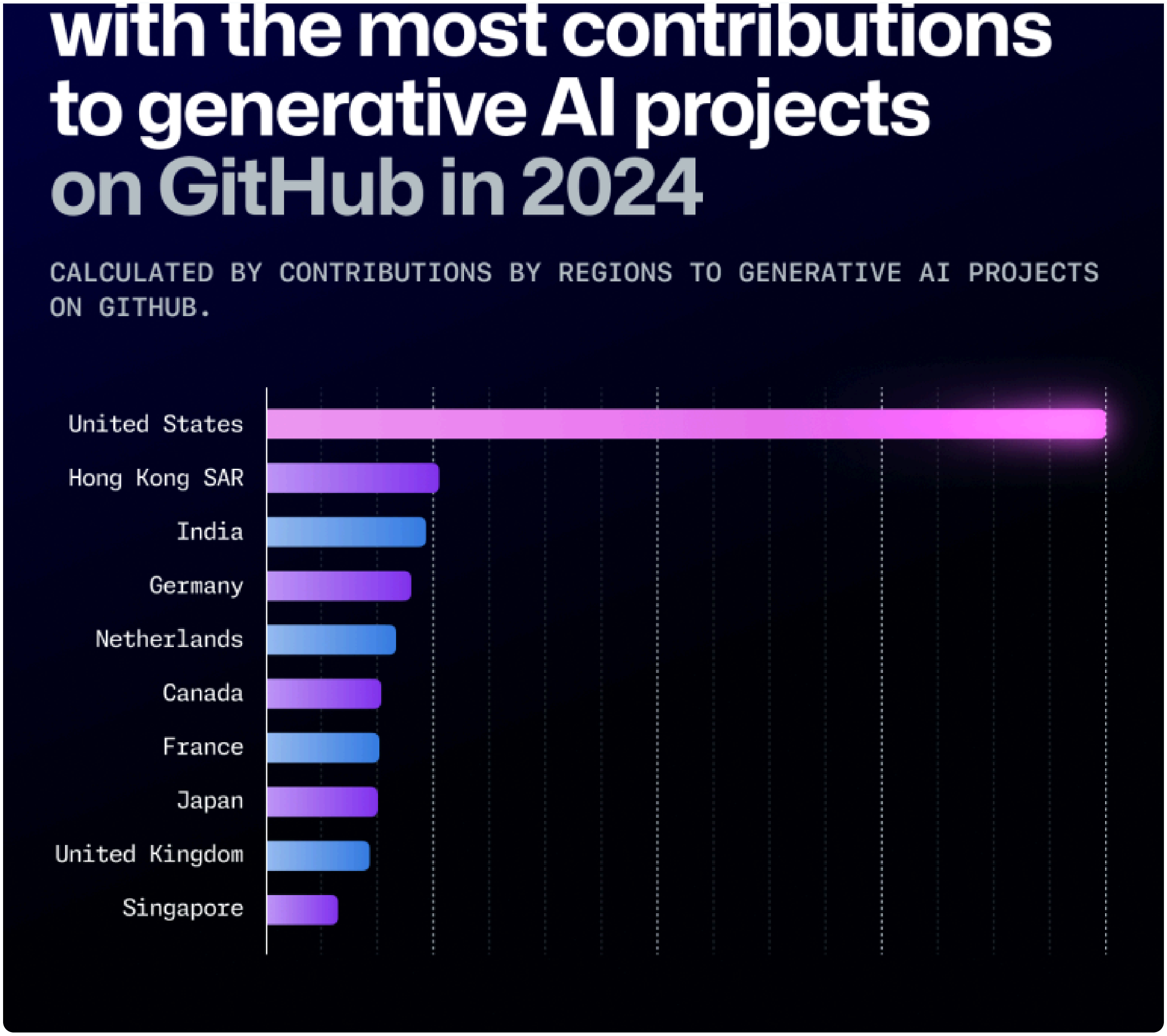
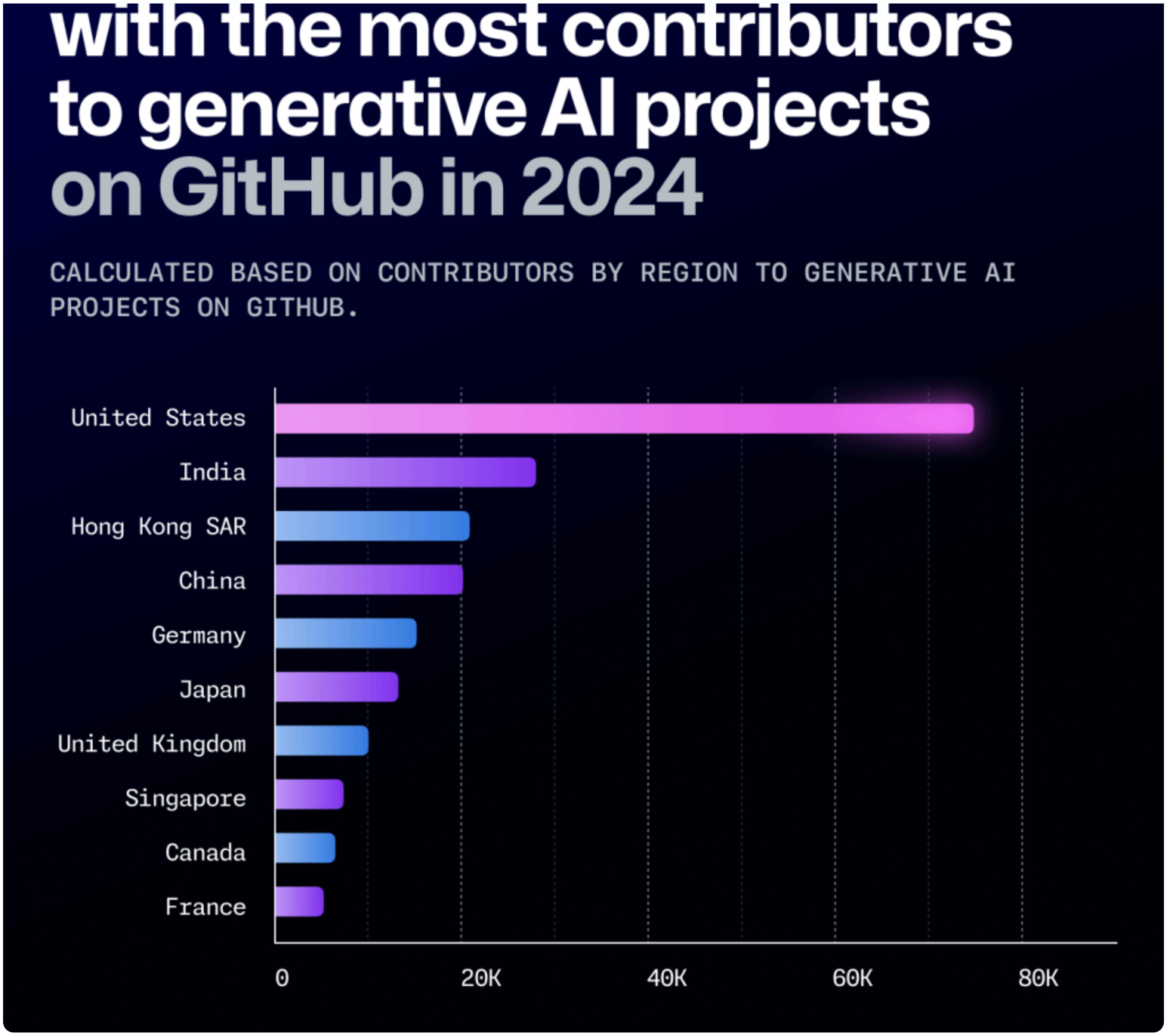

with the most contributions
to generative AI projects
on GitHub in 2024
CALCULATED BY CONTRIBUTIONS BY REGIONS TO GENERATIVE AI PROJECTS ON GITHUB.
United States
Hong Kong SAR
India
Germany
Netherlands
Canada
France
Japan
United Kingdom
Singapore



💡 **Stay smart.** When comparing regions with a high number of generative AI contributors versus regions with a high number of contributions, we see that while growth is still happening globally, the regions with larger developer populations are rising to the top.

🇮🇳 The next wave of software development in India

Behind only the United States in terms of developer population, India has the second-highest number of GitHub Education users, second-highest number of contributors to public generative AI projects on GitHub, and the second-highest number of

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The state of security and automation in 2024

In 2024, developers across GitHub used secret scanning to detect more than 39 million secret leaks. We also saw developers and open-source communities respond more quickly to security incidents through new generative AI security tools, automated alerts, and proactive measures. This isn't just helping make software more secure—it's leading to faster fixes, too.

The most common security vulnerabilities in 2024. `Injection` , an admittedly large category of security issues, was the most common type of vulnerability found across public and private repositories via CodeQL, a code analysis engine developed by GitHub to automate security checks. Meanwhile, `Security Logging and Monitoring Failures` vulnerabilities were found more often in private repositories.

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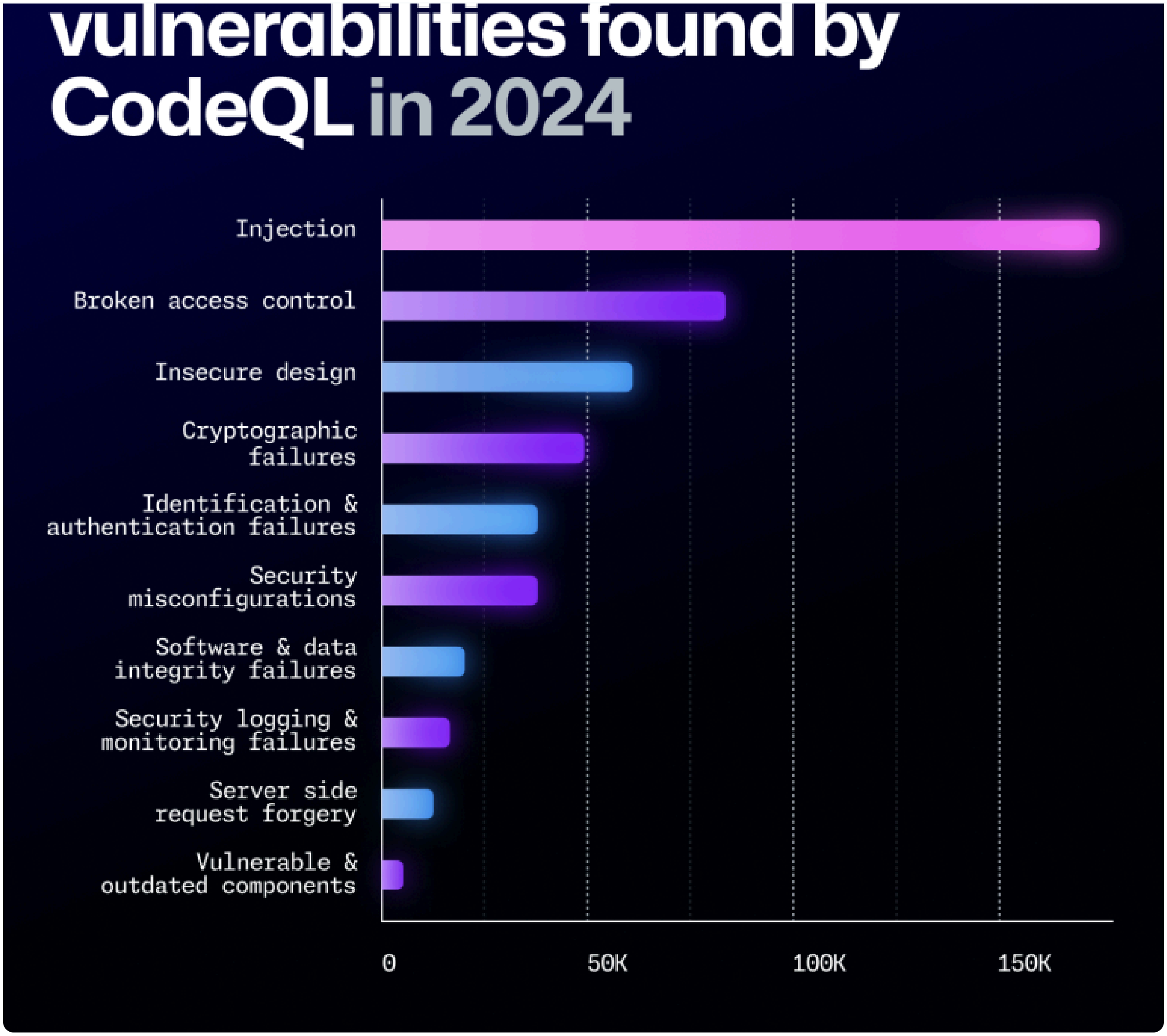


Developers are increasingly using AI for code reviews and security vulnerability remediation. AI doesn't replace security experts, but it can augment their knowledge and capabilities while helping address [a global shortage of security professionals](#).

- Notably, developers are experimenting with AI tools like [Copilot Autofix](#), an AI-powered security tool that automatically detects vulnerabilities and suggests fixes while offering explanations in natural language.
- We expect tools like this to improve security across open source and public projects—as well as with closed source, too. So far with Copilot Autofix, we've seen it helps developers:

- **Fix cross-site scripting vulnerabilities seven times faster,** reducing time to fix to 22 minutes, compared to almost three hours.
- **Fix SQL injection vulnerabilities twelve times faster**, cutting time to fix to just 18 minutes, compared to 3.7 hours.

Developers on GitHub are using automation to manage increasing security responsibilities. For instance, developers are merging an increasing number of pull requests generated by Dependabot, which sends alerts about outdated or vulnerable dependencies in a pull request. The gap between pull requests opened by Dependabot and pull requests merged by developers continues to shrink year over year, too.

While developers are using automation and AI to secure their code and applications, there's room to improve. Government regulations increasingly demand developers know the ingredients going into their software artifacts, which increases demand for implementing tools that automate governance and compliance.

94% of the top 50 open source projects are using the OpenSSF Scorecard to help ensure their projects implement security best practices. We evaluated this by looking at roughly 1 million repositories that have **OpenSSF scorecard**s in place from the top 50 most popular open source projects. The **OpenSSF Scorecard action** assesses repositories, runs checks for security best practices, and generates a security scorecard with real-time feedback.

Becoming familiar with **GitHub security features**, such as code scanning and secret scanning (which are free for open source developers), and supply chain governance features like **artifact attestations** is a good first step towards automating best security practices. Enterprise developers can also turn to their OSPOs for support in navigating regulations and implementing security measures across their open source dependencies, as **OSPOs will play increasingly critical roles in compliance**.

Developers are increasingly automating more aspects of build, test, and security activities using GitHub Actions in public and open source projects. In 2024, we saw developers use **10.54 billion total GitHub Actions minutes (measured in CPU minutes)**. That's up almost 30% year over year from the 7.3 billion GitHub Actions minutes developers used in 2023.

/ Blog

> "We get everything we need from the GitHub Actions marketplace to build and support our tailored CI/CD pipeline." – **Bjoern Bengelsdorf** // Senior Software Engineer, Otto Group

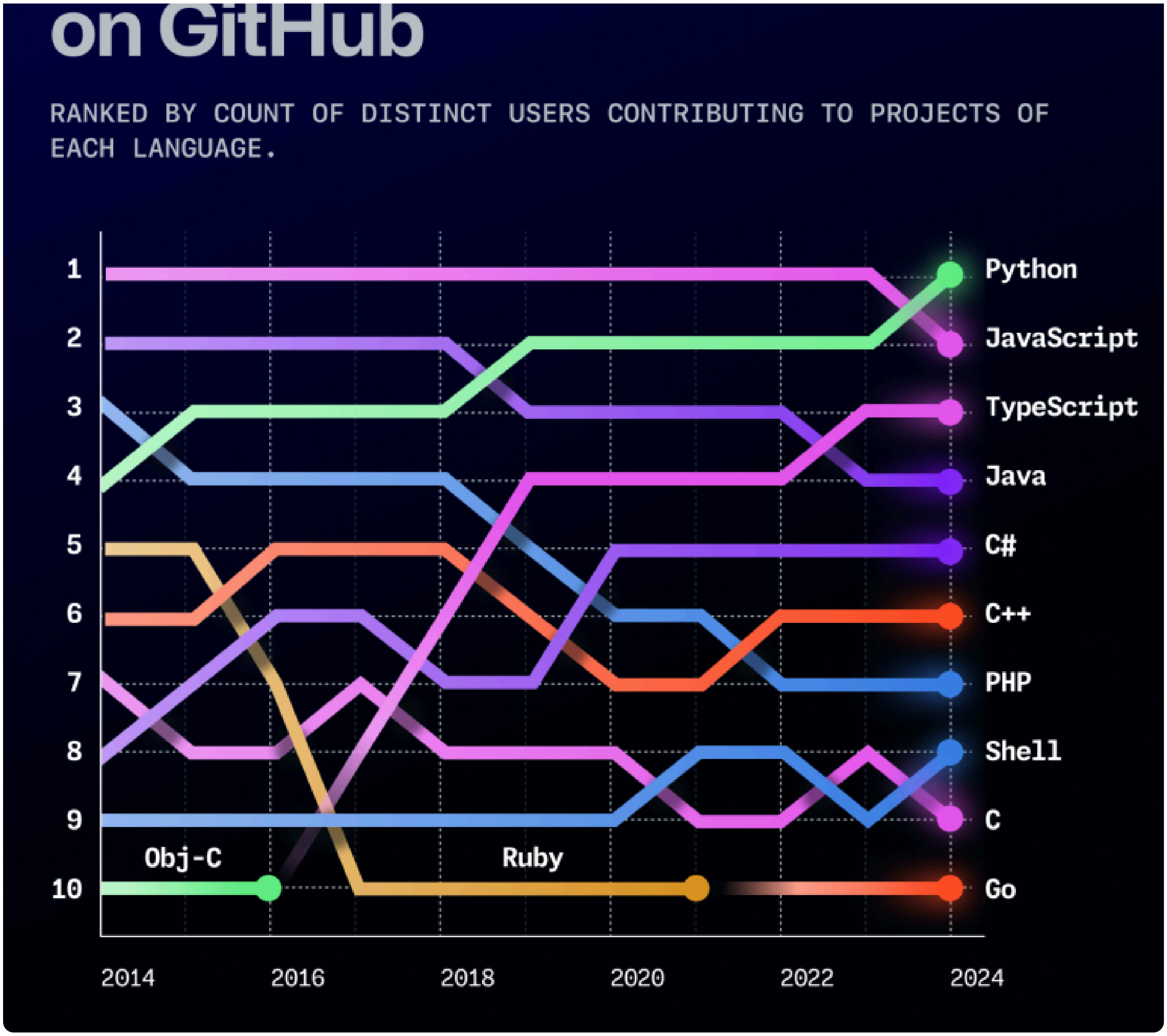
Among the most popular GitHub Actions in the GitHub Marketplace are OpenCommit, which augments commit messages with meaningful AI-generated content when pushing to remote, and Replexica, which provides AI-powered code translations across multiple programming languages. **These actions suggest that developers are finding more use cases for generative AI in their workflows.**

Check out the GitHub Marketplace and make your own GitHub Actions.

The most popular programming languages

generative AI boom we've seen over the past two years.

- **What the Python Software Foundation says:** We reached out to the Python Software Foundation, and Deb Nicholson, the foundation's executive director, gave us the following response, "Our goal is for Python to be a great tool that helps the ever-growing developer community build the world they envision. We couldn't be more pleased to learn about Python's continued rise in popularity on GitHub, especially coupled with the increased use of Jupyter Notebooks, data analysis, AI, and open source technology."
- **What else we're seeing:** Shell also overtook C in 2024. Though languages like Rust and Go are on the rise, more conventional languages are still heavily used and in demand. Additionally, high adoption of beginner-friendly languages like **JavaScript and Python** raises the possibility of more people learning how to code, as these are popular languages in settings like academia and data science.



How do we calculate the most used languages on GitHub?

We look at the totality of activity across commits, issues, pull requests, comments on issues and pull requests, discussions, pushed code, and reviewed pull requests, among other things. This helps us assess the diversity of actions of a given community across projects.

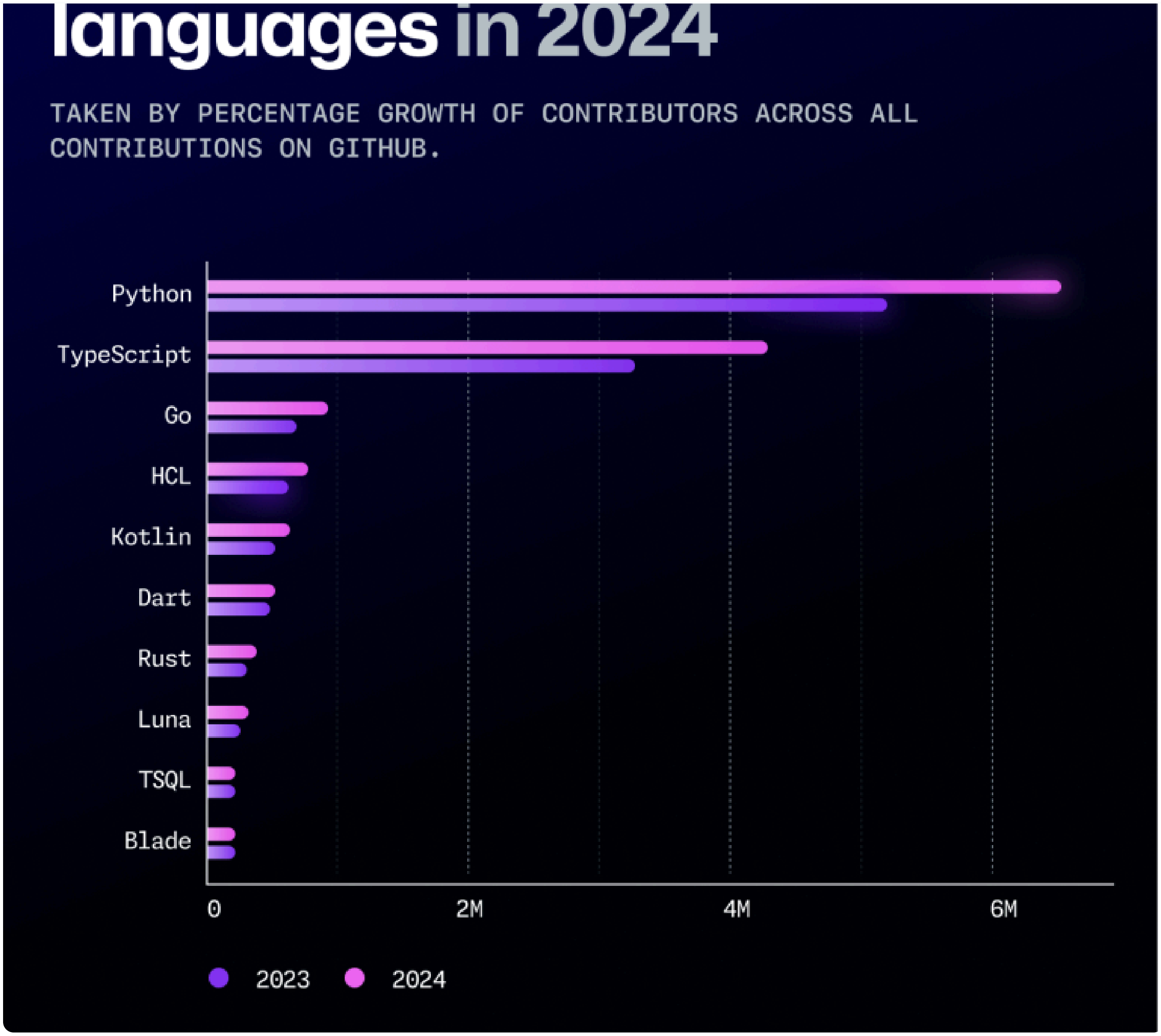
Notably, JavaScript still ranks first for code pushes alone. More developers still use JavaScript more often to push code, but in absolute activity across all contribution types on

- **TypeScript is cutting into JavaScript.** After growing exponentially from 2014-2019, TypeScript overtook Java last year to enter the top three programming languages on GitHub—and its continued growth speaks to its utility as a language, type checker, and compiler all in one. While Python is increasing in contributor counts for both code push activity alone and other activity faster than JavaScript, it isn't increasing in those faster than JavaScript and TypeScript combined. Rather than a slow down in the JavaScript community, what we are seeing is [a transition to TypeScript] for a large proportion of new commits. TypeScript is a superset of JavaScript and in the same npm ecosystem as JavaScript which makes it simple for JavaScript developers to gradually adopt.
- **JavaScript still maintains a massive developer base as we see increases in npm package consumption.** The language is versatile in running on both client and server sides, and easily adapts to different frameworks and standards, among other reasons for its popularity. And as its robust ecosystem continues to mature, we're seeing strong growth in the consumption of packages via the npm registry with a 15% year-over-year increase.

Explore the data yourself on the GitHub Innovation Graph 🔍

Last year, we launched the GitHub Innovation Graph which offers publicly accessible GitHub data for developers, researchers, and policymakers. We base our language rankings in the Innovation Graph on code pushes—so, JavaScript still holds the top spot.

See the data >


languages in 2024
TAKEN BY PERCENTAGE GROWTH OF CONTRIBUTORS ACROSS ALL CONTRIBUTIONS ON GITHUB.
Python
TypeScript
Go
HCL
Kotlin
Dart
Rust
Luna
TSQL
Blade
0
2M
4M
6M
2023
2024

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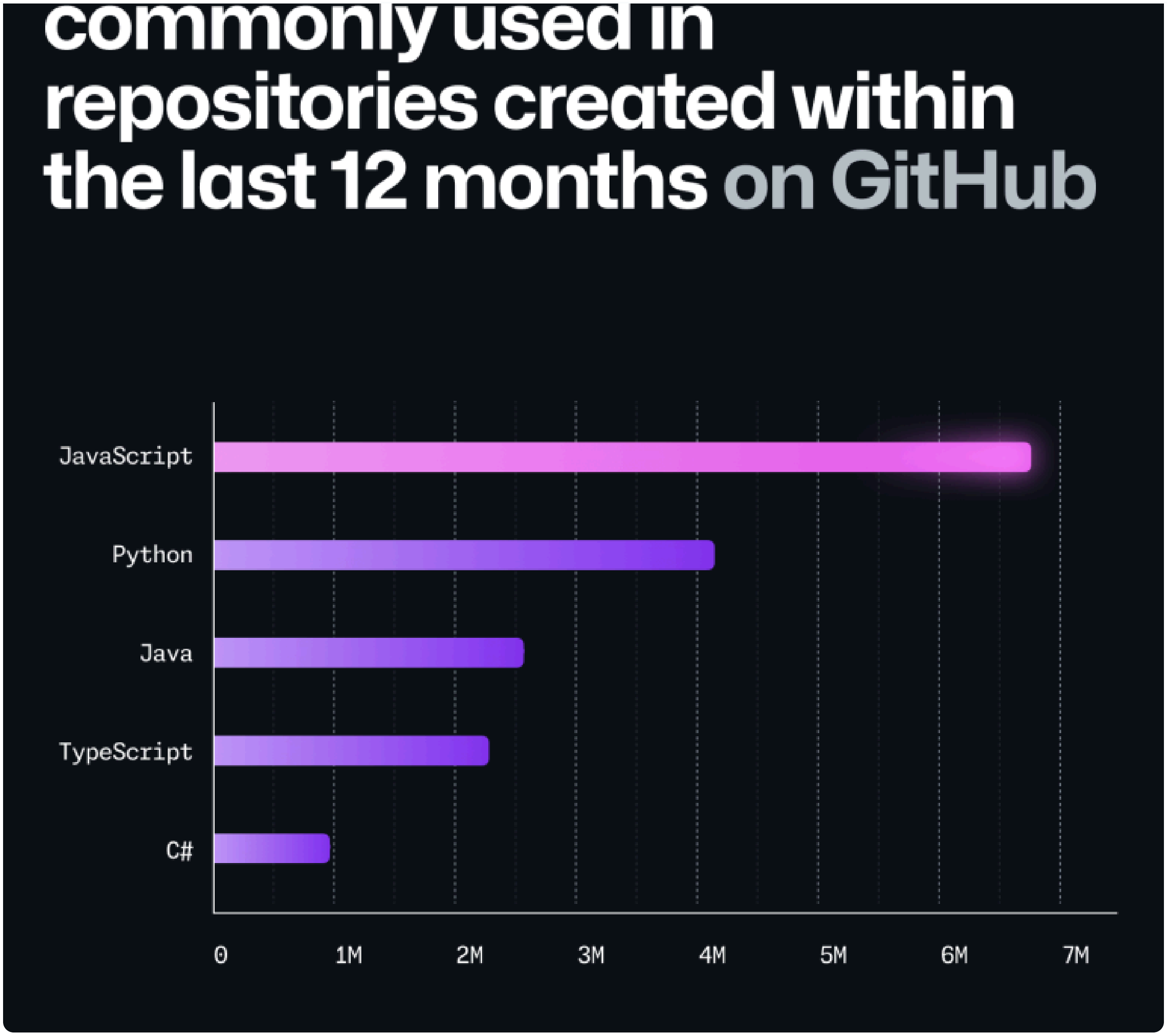

Rust continues to gain popularity for its safety, performance, and productivity. Originally intended to serve as a safer alternative to C and C++, Rust has exploded in popularity and adoption, with top applications, such as [Microsoft Windows](), using Rust to rewrite core libraries with its memory-safe code.

🦀 Learn how Rust developers are making the web safer and why Rust is the most admired language among developers.

/ Blog

- **Python** is the top preferred language for data science and research, and its continued growth over the past few years—alongside that of Jupyter Notebooks—may suggest that activity on GitHub is going beyond traditional software development.
- **T-SQL**, an extension of SQL that's primarily used within Microsoft SQL server, also indicates activity among data scientists and database administrators.

Read more on why Python keeps growing >

Researchers are an important and active community on GitHub

To support this community, GitHub partnered with ORCID in March 2024, the Open Researcher and Contributor ID nonprofit organization that provides a unique digital identifier for researchers. Researchers using GitHub can now link their ORCID iD to their GitHub profile so that their coding contributions are attributed to them, and tracked and visible to research and academic communities. We also introduced CITATION file support several years ago in a bid to help researchers on GitHub.

The continued popularity of HCL and Go reflect growth in operations and IaC work, particularly around managing cloud-native infrastructure. Since we first saw massive growth in cloud-native development in 2019, IaC has continued to grow in open source. The 25% year-over-year growth of HCL in particular suggests developers increasingly use declarative languages to dictate how they're managing cloud deployments.

The popularity of HCL and Go as well as Dockerfiles suggest that developers are scaling work in cloud-native applications. Increased Terraform use follows the increased use we've seen in Dockerfiles and other cloud-native technologies over the last decade. The increased adoption of IaC practices also suggests developers are bringing more standardization to cloud deployments.

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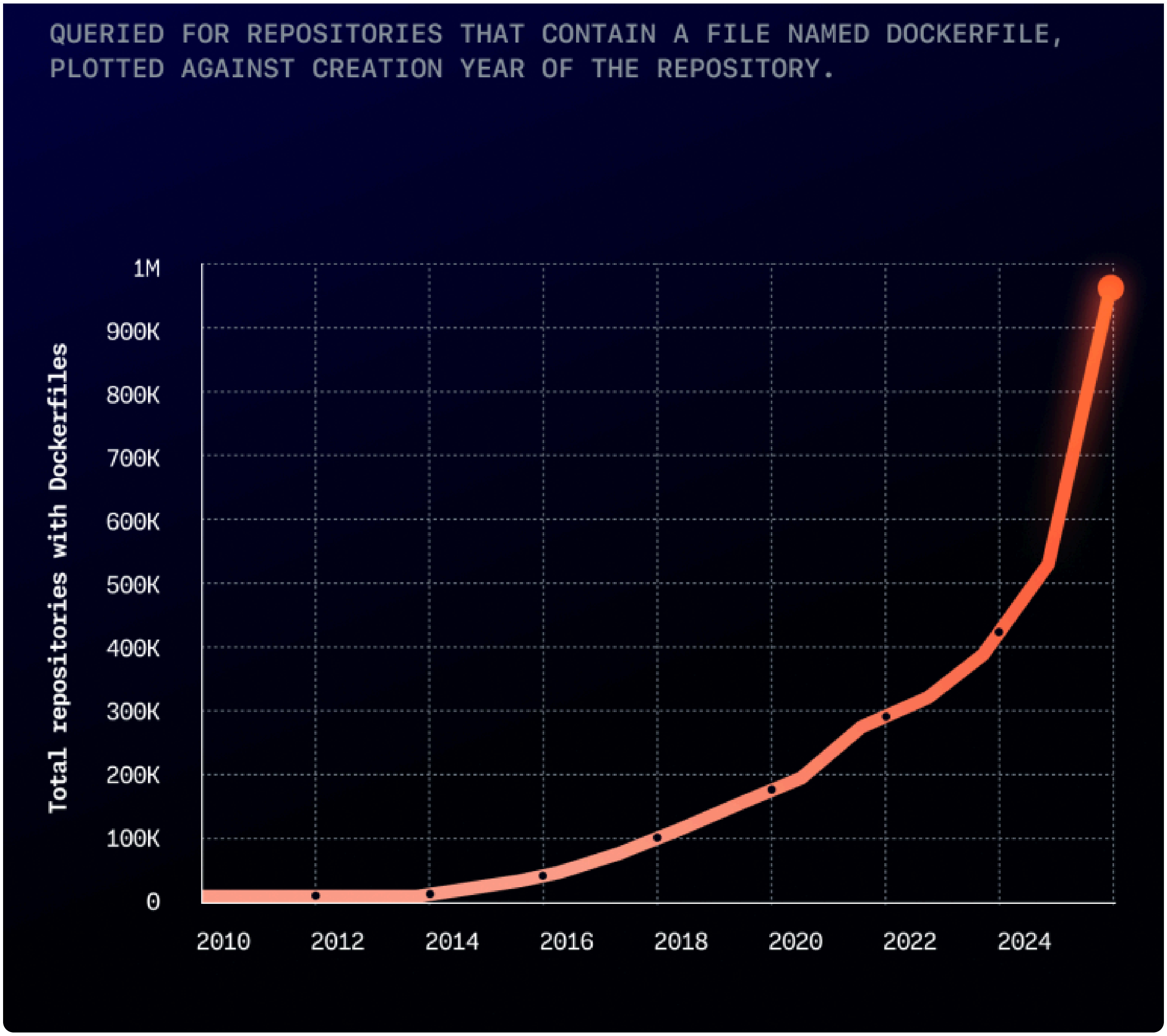

How we built data residency into GitHub Enterprise

This year, we introduced data residency to GitHub Enterprise that allows organizations to store their GitHub code in their preferred geographical region. And there were some interesting technical challenges when we built it—which may just surprise you.

[Learn more >](#)

As the developer's tech stack evolves, so does their role over time. We leave you with three takeaways:

- **Generative AI models are becoming core building blocks in software development**. They power coding tools that offer fixes and context behind vulnerability remediations, suggestions in response to natural language prompts and existing code, and facilitate learning among new and experienced developers alike. They're also changing how developers build applications, and developers will benefit from [platforms that allow them to easily experiment with AI models as building blocks]() without requiring separate setups or extra costs.
- **The global community of developers on GitHub is expanding rapidly—and the next generation of developers is getting started on GitHub.** An increasingly diverse community of developers drives innovation and talent, and refreshes the pool of solutions to increasingly complex problems. Increased access to and experimentation with AI could also simplify and personalize the coding journey for new developers, lowering entry barriers and further diversifying GitHub's community of developers.
- **The notion of who a developer is and the scope of what a developer does is changing**. The rise in Python, HCL, and Jupyter Notebooks, among other things indicates that the notion of a developer extends beyond software developers to roles like operations or IT developers, machine learning researchers, and data scientists.

Glossary

- **2024**: Refers to October 1, 2023 through September 30, 2024.
- **Contributions:** Commenting on a commit, issue, pull request, pull request diff, or team discussion; creating a gist, issue, pull request, or team discussion; pushing commits to a project; and reviewing a pull request.
- **Contributors**: GitHub users who have performed any of the contribution activities defined above
- **Developer:** Anyone with a GitHub account. Also sometimes referred to as a GitHub user. The open source and developer communities are an increasingly diverse and global group of people who tinker with code, make non-code contributions, conduct scientific

Generative AI: To find generative AI projects, we sourced a list of topic strings from GitHub CEO Thomas Dohmke's white paper with Keystone and searched for projects tagged with at least one of those topics.

- **GitHub Classroom users**: Anyone who has logged into GitHub Classroom. All GitHub Classroom users are GitHub Education users, but not all education users are classroom users.
- **GitHub Education user:** Includes GitHub Classroom users, couponed users (students, teachers), and users affiliated with an education organization.

Methodology

This report draws on anonymized user and product data taken from GitHub from October 1, 2023 through September 30, 2024.

More data is publicly available on the GitHub Innovation Graph—a research tool GitHub offers for organizations and individuals curious about the state of software development across GitHub. Only public activity is included, and metrics for economies are only reported when there are 100 or more unique developers performing the relevant activity within the time period.

For a complete methodology, please contact press@github.com.

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GitHub is the world's best developer experience and the only AI-powered platform with security incorporated into every step, so you can innovate with confidence.

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Let's dig into the 2025's open source data on GitHub to see what we can learn about the future.

Dylan Birtolo

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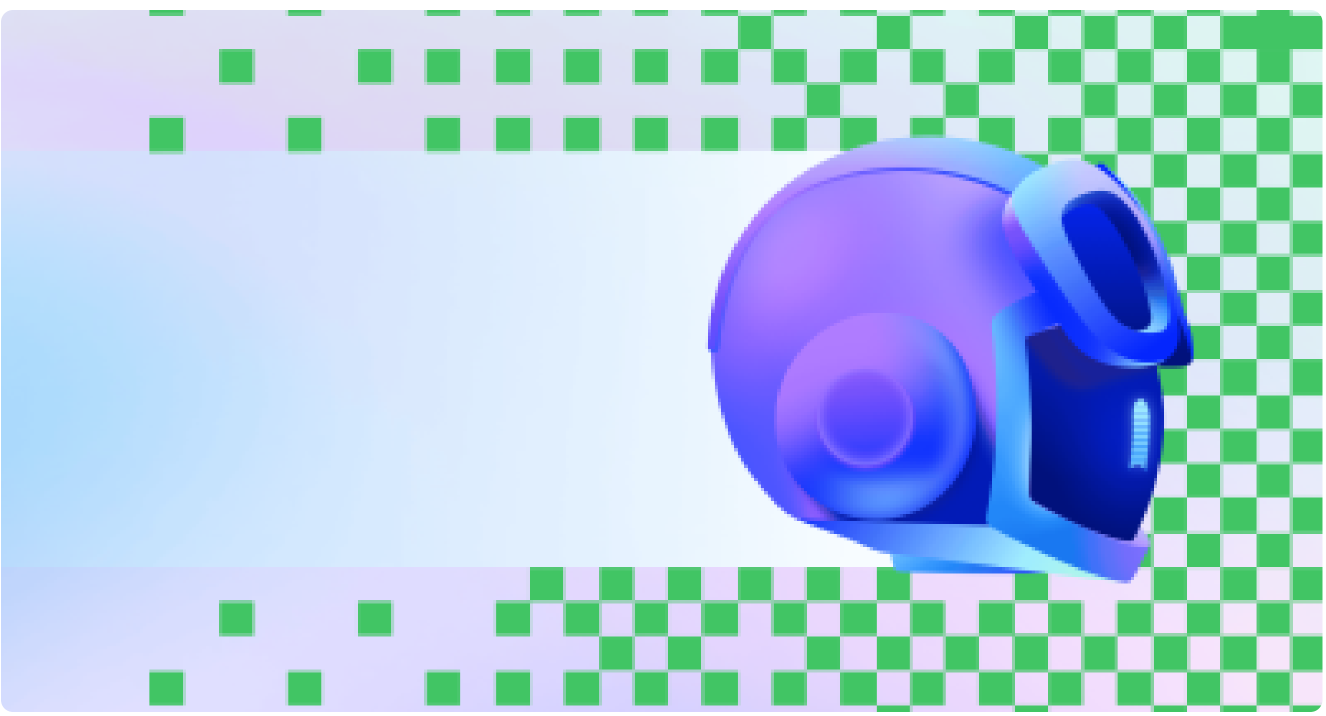
The ESC collection lets you escape the confines of your desk and get out into the sun where good ideas are bound to happen.

Lavinia Sfetcu



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